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Algorithm for Automatic Detection of Surface Atoms

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Automated identification of surface atoms is very convenient when, for instance, finding atoms that may desorb from a catalyst surface. The proposed algorithm for automated identification is based on the geometry of atom positions and quantifies the solid angle of "open space" around an atom. The solid angle is 2π sr for a prototypical surface atom, while the angle would be larger than 2π sr for a step edge atom, slightly larger than π sr for a surface atom at the foot of a step, and much smaller than π sr for a subsurface atom. The algorithm is expected to accelerate analysis of surface defects of slabs and nanoparticles and contribute to, for example, catalyst design.
Key words: Surface atoms, algorithm, automated detection, solid angle, convex hull, spherical triangle

1. INTRODUCTION

Surface point defects of metal oxides, such as O vacancies, have a dominant effect on heterogeneous catalysis [1-3]. As an example, the Mars-Van Krevelen mechanism is a prototypical catalytic process where the reaction sites are surface O vacancies on a metal oxide catalyst [4-7]. Step edge sites are less coordinated than terrace sites and often acts as active reaction sites in catalysts [8-10]. Quickly identifying surface and step edge atoms would assist efficient computational investigation of reactive sites. Visual inspection and identification of surface atoms take time, especially when attempting to evaluate more than tens of surfaces, and some sites could be easily overlooked in a complicated surface.

The coordination number is a quantity that represents the local environment, and a smaller coordination number than what is expected in bulk implies that the atom has missing bonds. One problem of the coordination number is that a clear-cut definition is difficult in some cases because the bond length cutoff is not unambiguous. Atoms in a prototypical simple cubic structure is considered six-fold coordinated; there are six nearest neighbors (NN) with exactly the same length, and the distance to the second NN (2NN) is $\sqrt{2}$ times that to the NN. However, in hexagonal La_2O_3 with the A-type rare earth structure, the coordination of La is typically considered to be seven-fold but the bond lengths are not equal; there are four short bonds and three short bonds [11]. As a result, one can also claim that La is four-fold coordinated. Use of the (partial) pair correlation function, which describes the probability that an atom exists at a certain distance from the atom at the origin, removes the issue of arbitrariness in the bond distance cutoff. However, to judge that bonds are missing in a certain atom, some baseline pair correlation function is necessary, which needs an assumption that each atom corresponds to a certain site in a crystal. Moreover, the pair correlation function is a function of distance only and information on the direction (angles) has been stripped. Therefore, a reduced pair correlation function cannot distinguish whether nearby atoms were removed in random directions or to form a surface.

This paper proposes an algorithm to derive a quantitative value for an atom that describes the geometric environment regarding neighboring atoms. No underlying assumption regarding the original structure is necessary. The effectiveness of the proposed algorithm in identifying surface atoms is demonstrated in several simple and complex surfaces.

2. ALGORITHM

An atom in bulk has neighboring atoms in all directions around it, but an atom at a prototypical surface has neighboring atoms in half of the space around it. Therefore, a useful value that describes the geometric environment is the solid angle of the "open space" around the atoms, which is calculated using the following procedure.

1) For an atom i with Cartesian coordinates (x_i, y_i, z_i), find a list of all atoms j within an arbitrarily determined cutoff distance d from atom i; the Cartesian coordinates of atom j is denoted as (x_j, y_j, z_j). The number of such atoms j is denoted as J.
2) Normalize the distance between atom j and atom i. The following vectors are defined.
$$\vec{R}_j = \left(x_j - x_i, y_j - y_i, z_j - z_i\right) \text{ and } \vec{r}_j = \vec{R}_j \big/ \left|\vec{R}_j\right|. \quad (1)$$
Note that $\vec{r}_j$ are unit vectors.
3) Remove duplicate vectors among $\vec{r}_j$.
4) Obtain the negative of the average of $\vec{r}_j$, $\vec{R}'$, and the unit vector proportional to it, $\vec{r}'$. These are defined as
$$\vec{R}' = -\sum_{j=1}^{J} \vec{r}_j \text{ and } \vec{r}' = \vec{R}' \big/ \left|\vec{R}'\right|. \quad (2)$$
If $\vec{r}'$ is a zero vector (in practice, if $\left|\vec{R}'\right|$ is sufficiently short), further analysis is not possible.
5) $J \geq 3$ must hold and all j must not be on a single plane containing the origin for step 5; otherwise go to step 6 (J=1) or 7 (other cases). Obtain the convex hull of points represented by unit vectors $\vec{r}'$ and $\vec{r}_j$. This can be accomplished by, for example, using the qhull code [12]. All of these points are on the unit sphere and

therefore appear on the convex hull. All facets where $\vec{r}'$ is a vertex should be triangles; if not, degeneracy need to be removed by slightly displacing a number of atoms. Consider these facets as spherical triangles, and obtain the sum of the area (spherical excess) of all such spherical triangles. The area S of the spherical triangle with corners defined by unit vectors $\vec{r}'$, $\vec{r}_p$ and $\vec{r}_q$ is (Eq. (1.8) of Ref. [13])

$$S = 2\arccos\frac{\left(1+\vec{r}'\cdot\vec{r}_p+\vec{r}'\cdot\vec{r}_q+\vec{r}_p\cdot\vec{r}_q\right)}{\sqrt{2\left(1+\vec{r}'\cdot\vec{r}_p\right)\left(1+\vec{r}'\cdot\vec{r}_q\right)\left(1+\vec{r}_p\cdot\vec{r}_q\right)}}. \quad (3)$$

The unit of S is steradians (sr) because S can be viewed as a solid angle.

S may be obtained using the atan2 function implemented in many computer languages by applying the relation

$$\arccos x = 2\arctan\sqrt{\frac{1-x}{1+x}}. \quad (4)$$

The denominator is set to be positive to choose the correct branch.

6) If J=1, atom i is bonded to a single atom, thus all space around atom i is considered empty and S=4π sr.
7) If $J \geq 2$ and all j are on a single plane containing the origin, all j are on a unit circle with the origin at the center. Taking the length of the longest arc between neighboring j as θ where $2\pi/J < \theta < 2\pi$ and $S = 2\theta$. Details are given in section 4.2.
8) Whether the atom is at a surface is determined using the value of S. In general, S is much larger than 2π for a step edge atom when considering steps on a surface, roughly 2π for a terrace atom, slightly larger than π for a surface atom at the foot of a step, and much smaller than π for a subsurface atom. $S > 0.75\pi$ sr appears to be a reasonable criterion for a surface site (details are given in the following section).

We stress that the above algorithm does not make any assumption on how atoms are arranged, including the underlying crystal structure; it can be used for slabs, nanoparticles, or even bulk. The algorithm is applied to specific examples in section 3. The proposed algorithm is first demonstrated on a smooth surface, namely the (111) surface of a fcc (face centered cubic) structure. Applications to rough surfaces, for example surfaces containing steps, are then considered to test the behavior of the algorithm.

3. COMPUTATIONAL METHODS

The algorithm to obtain nonpolar slab-and-vacuum models derived by Hinuma et al. [14, 15] was employed in this work. These models were then relaxed using first principles calculations for sections 4.2, 4.4, and 4.5; the projector augmented-wave method [16] as implemented in the VASP code [17, 18] and the Perdew-Burke-Ernzerhof functional tuned for solids (PBEsol) [19] within the generalized gradient approximation (GGA) was used. An effective Hubbard U [20] of 3eV was added on 3d states of Ti in sections 4.2 and 4.5.

4. APPLICATIONS AND DISCUSSION

4.1. (111) surface of the fcc structure

The algorithm is applied to the outermost surface atom of the (111) surface of a fcc structure as a simple example. Fig. 1(a) shows a surface atom (red ball) and NN atoms (green balls). Atoms in a fcc structure have 12 NN atoms in bulk, but three are missing at this surface. The NN distance has been normalized to unity such that $\vec{r}_j = \vec{R}_j$. The sum of $\vec{r}_j$ over the topmost layer, $\sum_{j=1}^{6}\vec{r}_j$, is zero, hence $\vec{r}' = -\frac{1}{\sqrt{6}}\sum_{j=7}^{9}\vec{r}_j$ [shown as a blue ball in Fig. 1(b)]. The set of vertices of facets of the convex hull containing $\vec{r}'$ are $\{\vec{r}',\vec{r}_1,\vec{r}_2\}$, $\{\vec{r}',\vec{r}_2,\vec{r}_3\}$, ... $\{\vec{r}',\vec{r}_6,\vec{r}_1\}$. Considering these facets as spherical triangles [one of them is shown in yellow in Fig. 1(b)], the area of a spherical triangle is exactly 1/12 of the area of the sphere, hence the sum of the area of the six spherical triangles is 2π. In other words, the solid angle of the "open space" is 2π sr, which matches intuition because there are no atoms over a hemisphere. Fig. 1(c) shows the case when atoms up to 3NN is considered. The $\vec{R}_j$ of second nearest neighbor (2NN) and 3NN atoms are divided by $\sqrt{2}$ and $\sqrt{3}$, respectively, to obtain $\vec{r}_j$ [small balls in Figs. 1(d-e)]. In this case too, the open space is a hemisphere and the solid angle is 2π sr [Fig. 1(f)].

4.2. (110) surface of rutile TiO_2

The (110) surface of rutile TiO_2 from two directions is shown in Fig. 2(a,b). The outermost atom is O (one representative is labeled as "A" in Fig. 2) that is bonded to two Ti ("B" in Fig. 2). If d is taken as the NN distance, which is normalized as unity, these two Ti lie on a unit circle centered at A [Fig. 2(c)]. Here, $\vec{r}_1 = \vec{R}_1$ and $\vec{r}_2 = \vec{R}_2$. There are two arcs between neighboring j with lengths θ_{12} and θ_{21}, which are the minor and major arcs, respectively. Note that the arc length and central angle are identical in a unit circle. According to step 7 of the algorithm, θ is chosen as the larger of the two ($\theta = 1.41\pi$ rad). The unit circle in Fig. 2(c) is then viewed as a great circle of a unit sphere [circle C_0 in Fig. 2(d)]. The two ellipses in Fig. 2(d) represent great circles, which are labeled as C_1 and C_2, that are perpendicular to the great circle C_0. A reasonable definition of the open space around A is the solid angle corresponding to the part of the sphere shown in blue in Fig. 2(d), which is exactly $S = 2\theta$ sr. Fig. 2(e) is when d is taken as the 2NN distance; it is an example when $J > 2$ but all j are on a single plane. There are now three arcs between neighboring j, with lengths θ_{21}, θ_{13}, and, θ_{31}, but S does not change from $S = 2.81\pi$ sr. When d is taken as the 3NN distance, all j do not lie on the same plane, step 5 in the algorithm is employed, and $S = 3.01\pi$ sr. This value is larger than when d is the NN or 2NN distance ($S = 2.81\pi$ sr) because the latter assumes that hypothetical atoms at the intersections of C_1 and C_2, which are labeled as "E" in Fig. 2(d), are implicitly considered as additional NN atoms. These hypothetical atoms are on the plane defined by the outermost atoms, as is shown in Fig. 2(a), and removing these increases the open space around the atom labeled as "A".

4.3. (211) surface of the fcc structure

The (211) surface of the fcc structure [Fig. 3 (a)] is used as a surrogate surface to model under-coordinated step edge sites in fcc metals because this surface can be regarded to consist of {111} and {100} facets. There is one type of atom in each layer, and representative atoms in the first to seventh

layers are labeled as atoms 1-7. The NN to 3NN atoms for atoms 1 to 3 are shown in Figs. 3 (b-d), respectively.

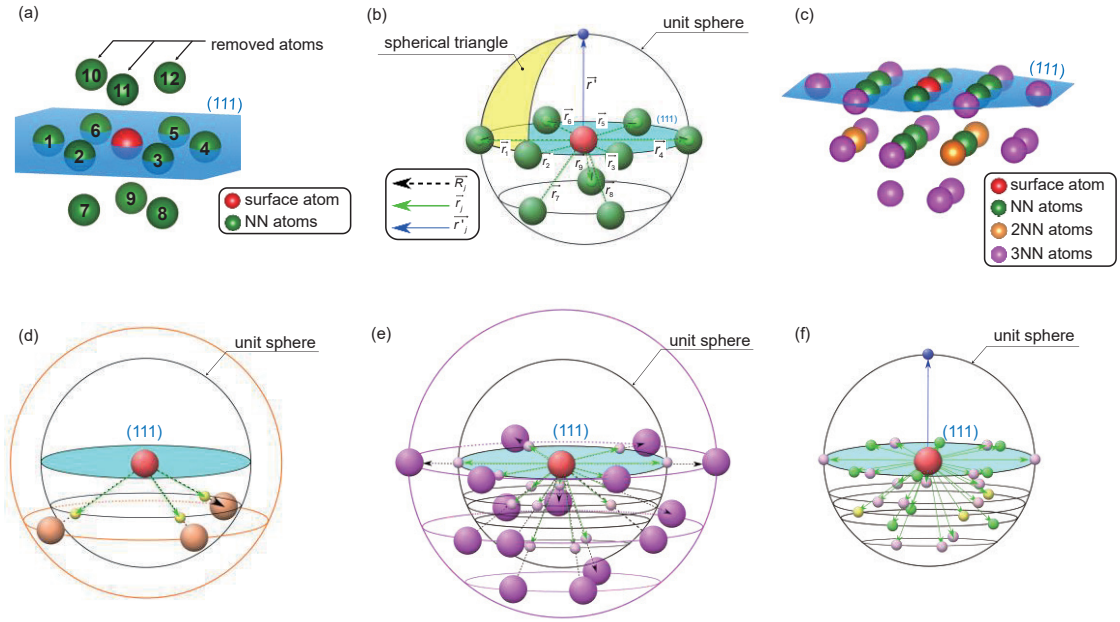

Fig. 1. Application of the algorithm to the (111) surface of a fcc structure. (a) Cleaving a (111) surface (blue plane). There are 12 NN atoms (green balls) to the surface atom of interest (red ball), and three are removed by cleaving. (b) NN atoms are shown on a unit sphere centered at the origin. The vectors $\vec{r}_j$ and $\vec{r}'$, as well as a spherical triangle with vertices $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}'$, are shown. (c) NN, 2NN, and 3NN atoms to a (111) surface atom. (d) Projections of the 2NN atoms (large orange atoms) to the unit sphere (small yellow balls). (e) Projections of the 3NN atoms (large purple atoms) to the unit sphere (small pink balls). (f) Projections of the NN, 2NN, and 3NN atoms to the unit sphere (small green, yellow, and pink balls, respectively). The blue ball corresponding to $\vec{r}'$ is also shown.

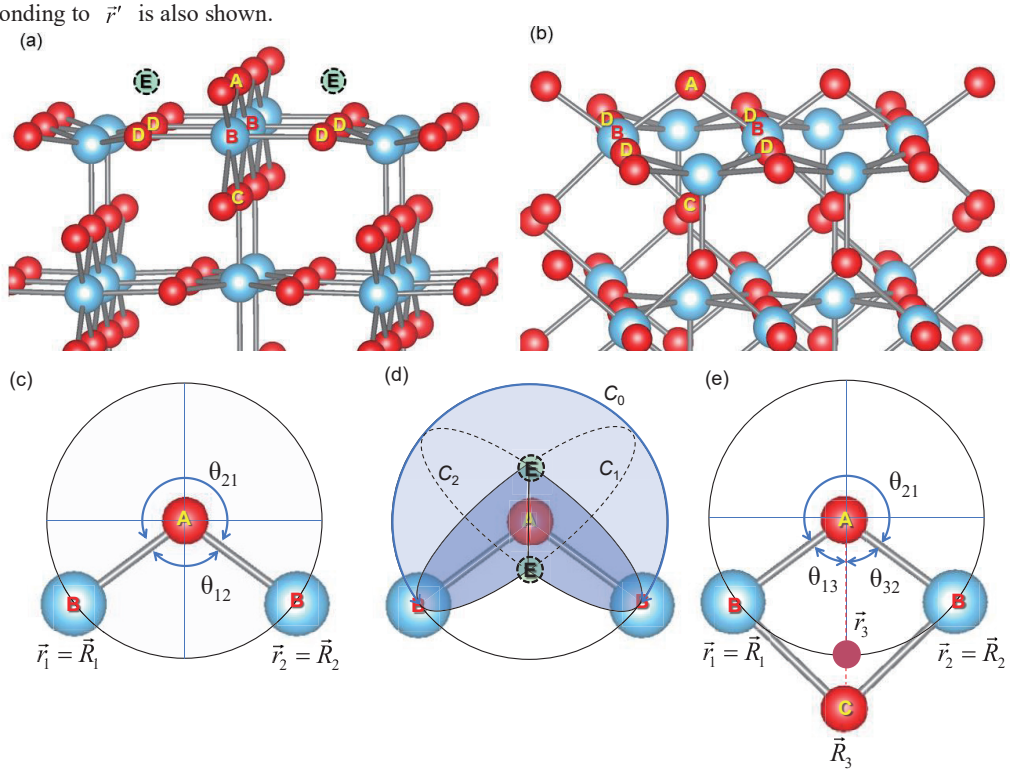

Fig. 2. Application of the algorithm to the the rutile TiO_2 (110) surface. (a,b) The surface shown from two directions. Blue and red balls correspond to Ti and O, respectively. The positions labeled as "E" represent hypothetical atoms. (c) Ti atoms labeled as "B" in (a,b) are shown on a unit circle with O atom "A" at the center. (d) Atoms "B" are positioned on a great circle in a unit sphere with "A" at the center. (e) Atom "C" and its corresponding position on the unit circle (pink point) is added to (c).

Fig. 3(e) plots S as a function of d for atoms 1-7. S of atom 1 exceeds 2π sr when d is small because this is a step edge atom and the open space is larger than a hemisphere. S gradually decreases to 2π sr when d is increased; atom 1 is a topmost surface atom when atoms on adjacent step edges are considered. On the other hand, S for atom 2 is 2π sr when $d \le \sqrt{2}$ (up to 2NN is considered) as this atom is on a terrace surface and cannot see any steps. S decreases when the 3NN is considered ($d = \sqrt{3}$) because the step edge is now visible. S for atom 3 is almost independent of d since this atom is at the foot of a step regardless of d. We note that the values of S for atoms 1 to 3 when d is the NN ($d = 1$) or 2NN distance ($d = \sqrt{2}$) is exactly what is intuitively expected using the angle between (100) and (111) orientations [Fig. 3(a)]. Atoms that are deeper (4 to 7) cannot see the surface at $d = 1$, thus $\vec{r}'$ is a zero vector and S is not defined. However, once the atoms can find missing atoms at the surface when d is increased, $\vec{r}'$ no longer becomes a zero vector and S can be obtained. Based on the above observations, S is much larger than 2π sr for a step edge atom, roughly 2π sr for a terrace atom, slightly larger than π sr for a surface atom at the foot of a step, and well below π sr for a subsurface atom.

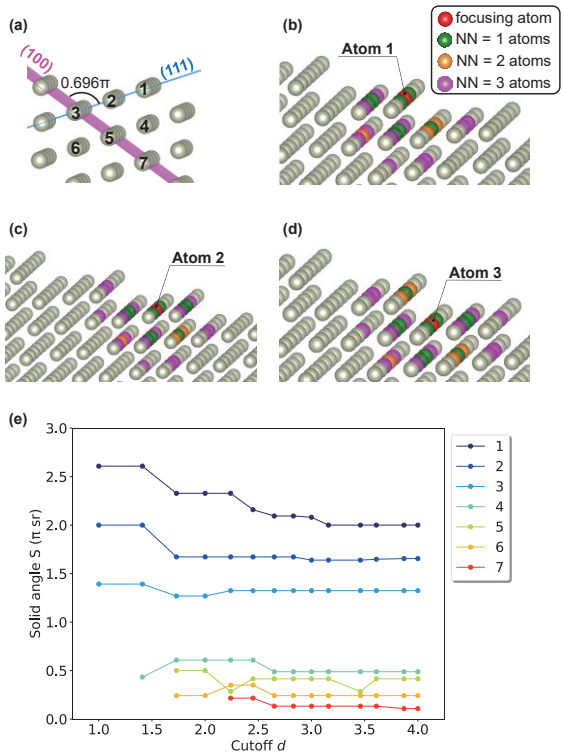

Fig. 3. Application of the algorithm to the (211) surface of a fcc structure. (a) A representative atom each in the first to seventh layer are shown (1-7) together with (100) and (111) planes. NN, 2NN, and 3NN atoms are shown for atoms in the (b) first, (c) second, and (d) third layers, respectively. (e) Solid angle S as a function of the cutoff d, in units of the NN distance, shown for the seven atoms in (a).

The effect of removing duplicate $\vec{r}_j$ in step 3 of the algorithm is reflected in Fig. 3(e). S when the cutoff is the 4NN and 14NN (d = 2 and 4) is the same as the 3NN and 13NN ($d = \sqrt{3}$, and $\sqrt{15}$), respectively, because the set of $\vec{r}_j$ from the 4NN and 14NN is the same as that from the NN (d = 1). In consequence, addition of these points does not affect S. Similarly, S when the cutoff is the 8NN ($d = 2\sqrt{2}$) is the same as when the cutoff is the 7NN ($d = \sqrt{7}$) because the set of $\vec{r}_j$ from the 8NN matches those of the 2NN ($d = \sqrt{2}$).

4.4. $(11\bar{3})$ surface of β-Ga_2O_3

β-Ga_2O_3 has low symmetry (space group symbol $C2/m$) and shows catalytic activity [21, 22]. Although the (100) surface has the lowest surface energy, the $(11\bar{3})$ surface is considered more reactive while being experimentally accessible; the minimum surface O vacancy formation energy of the $(11\bar{3})$ surface is smaller by slightly more than 1 eV than the (100) surface[23]. The minimum bond length of β-Ga_2O_3 is 1.86 Å. The $(11\bar{3})$ surface (Fig. 4) can be viewed as terraces and steps. In this viewpoint, the distance between step edges is 10.3 Å, thus S for atoms that exceed 2π sr is expected to gradually decrease to 2π sr when d is increased to above 10.3 Å. S for Ga atoms 1-9 and O atoms A-M in Fig. 4 is plotted as a function of d in Figs. 5(a) and 5(b), respectively. In general, the value of S becomes less dependent on d when d > 3.7 Å, which is roughly twice the nearest neighbor distance. Therefore, S for d > 3.7 Å is considered in the subsequent discussion.

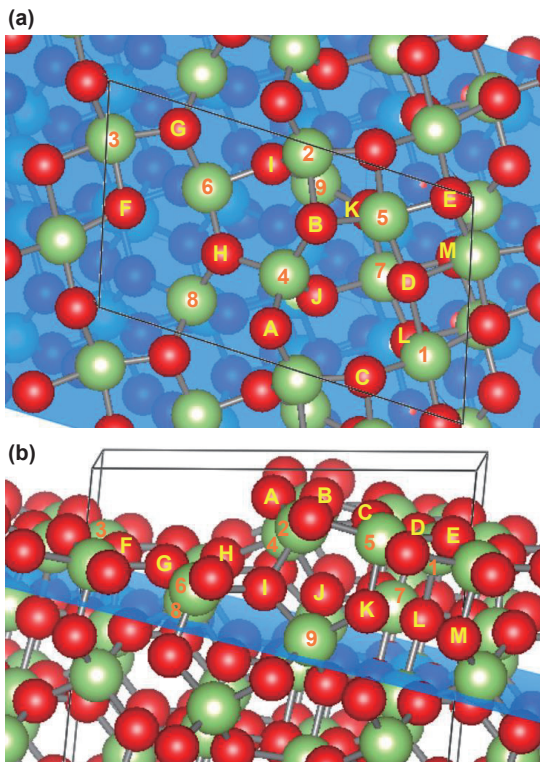

Fig. 4. The $(11\bar{3})$ surface of β-Ga_2O_3. Green and red balls represent Ga and O, respectively. The blue planes are positioned at the same location in panels (a) and (b).

S of Ga atoms are examined first. S of atoms 1-5 lie between 1.5π and 2.2π sr, suggesting that these are surface sites. S of atom 6 is around 1.2π to 1.3π sr, which is a reasonable value for a surface atom at the foot of a step. S of atom 8 is small at 0.8π to 0.9π sr. Although this atom is visible from the surface, it is at the foot of a step and its position is slightly away from the terrace surface compared

to other surface Ga. S of other Ga (atoms 7 and 9) are smaller than 0.5π sr.

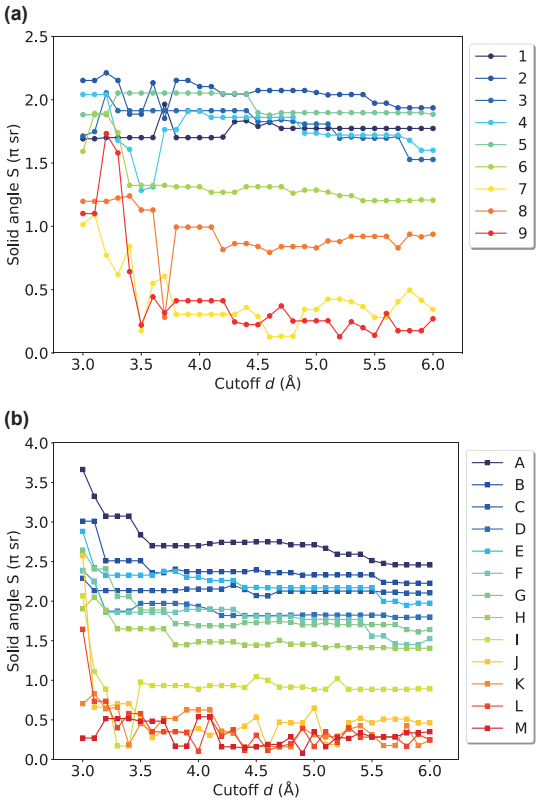

Fig. 5. S as a function of d for (a) Ga and (b) O atoms on the $(11\bar{3})$ surface of β-Ga_2O_3. The atom labels correspond to those in Fig. 4.

O atoms are discussed next. Atom A is a step edge atom and is the only two-fold coordinated atom. Its S exceeds 2.5π sr in the range of d in the figure. S of surface O atoms B-H range between 2.4π to 1.4π sr, where atoms closer to the step edge foot have a lower S. Atom I is a step edge site that is barely visible from the surface; its S is roughly 0.9π sr. Other atoms J-M cannot be considered surface sites and S is smaller than 0.6π sr. The observations suggest that S~0.75π sr can be used as a criterion for detection of surface sites, and step edge sites can have a very large S over 2.5π sr.

4.5. Step model of anatase TiO_2 with (101) terrace surface

A computer-assisted algorithm by Hinuma et al. to obtain slab-and-vacuum models can give a stoichiometric step edge model of the anatase TiO_2 (101) surface shown in Fig 6(a)[24]. However, some atoms must be removed to obtain the stoichiometric model in Ref. [9]. We attempt to use the proposed algorithm to find atoms that should be removed. The NN bond length for anatase TiO_2 is 1.97 Å in first principles calculations (VASP with PBEsol and effective Hubbard U [20] of 3eV on 3d states of Ti), and three values of d are considered, which are 4, 4.5, and 5 Å. S is extremely large in atom O1; the value is 3.28π sr for d = 4 and 4.5 Å and 3.18π sr for d = 5 Å, respectively. The next largest S is 0.8π sr smaller. An overhanging step edge atom can have S larger than 3π, which means than three-fourths of the sphere around the atom is open space. Therefore, atom O1 is removed in the first step, resulting in Fig. 6(b). The largest S in Fig. 6(b) is found in atom Ti1; its value is 3.00π sr for d = 4 Å and 2.76π sr for d = 4.5 and 5 Å. The next largest S for Ti is 1.61π sr for all three d, which is smaller by at least 1.1π sr, thus clearly Ti1 should be removed. The result is Fig. 6 (c). One more O must be removed to attain stoichiometry. There are three atoms with S > 2.40π sr for all three d, which are atoms O2 to O4. The next largest S is 1.81π, 2.24π, and 2.15π sr for d = 4, 4.5, and 5 Å, respectively. Figs. 6(d-f) shows the models where either O2, O3, or O4 was removed from Fig. 6(c). After relaxation, Fig. 5(e) converged to the structure starting from Fig. 6(d), and the energy of the structure from Fig. 6(f) had 3.8 eV/removed O higher than that from Fig. 6(d). The stability of the structure of Fig. 5(d) is very apparent by intuition; the one-fold coordinated O was removed in Fig. 6(d) but two-fold coordinated O was removed in Figs. 6 (e, f). The model in Ref. [9] is the same as Fig. 6 (d), which means that we successfully discovered this model by identifying and removing overhanging step edge atoms using S defined in this paper and considering information on the coordination.

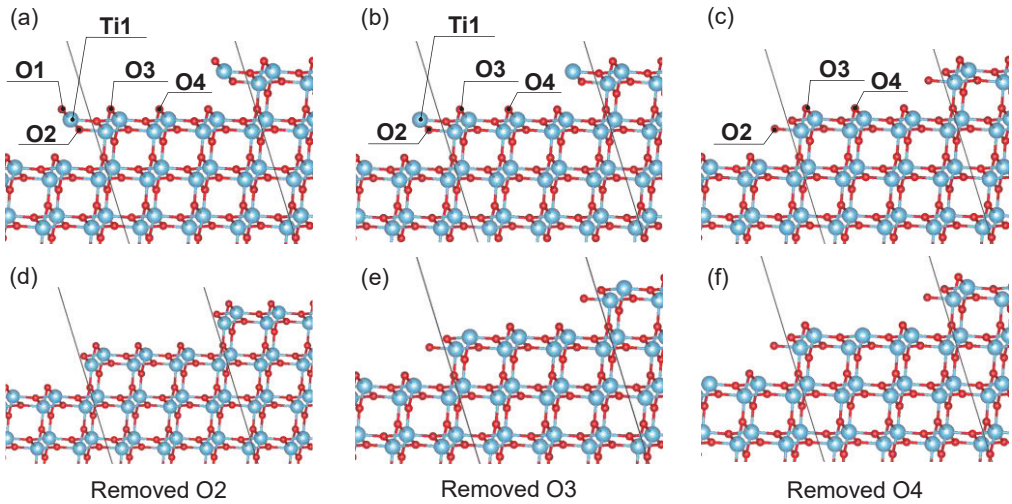

Fig. 6. Removing atoms to make a reasonable step model of anatase TiO_2 with the (101) terrace surface. Red and blue balls indicate Ti and O atoms, respectively.

5. CONCLUSIONS

An algorithm to identify surface sites based on geometry was derived and demonstrated. The open space around an atom, S, was quantitatively defined and expressed as a solid angle. A rule of thumb is that the solid angle of open space around each atom is much larger than 2π for a step edge atom when considering steps on a surface, roughly 2π for a terrace atom, slightly larger than π for a surface atom at the foot of a step, and much smaller than π for a subsurface atom. $S > 0.75\pi$ sr appears to be a reasonable criterion for a surface site. This type of surface atom identification would accelerate analysis of surface defects and contribute, for example, to catalyst design.

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